

Final Project Reflection

Name: Name intentionally omitted

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Objective:

1. Describe the dataset
2. Determine the Measures of Center in the dataset
3. Determine the Measures of Variation (Spread) in the dataset
4. Determine the Measures of Location in the dataset

After completing the math, I had to:

1. Transfer the dataset as an array in Visual Basic
2. Create a program that calculates the Measures of Center, Measures of Spread, and Measures of Position of the dataset

Verifiable Website (Lodging by month (excluding taxes) | October 2020 - September

2021): https://www.gsa.gov/travel/plan-book/per-diem-rates/per-diem-rates-lookup/?action=perdiems_report&state=OH&fiscal_year=2021&zip=&city=

Description: This project used data from the lodging rates by county in Ohio from October 2020 to September 2021, but specifically from March 2021.

Data Analysis:

Lodging by month (excluding taxes) | October 2020 - September 2021 (Raw Data)

Per Diem Lodging Rates from March 2021 (x)

\$99
\$96
\$151
\$137
\$122
\$109
\$117
\$105
\$101
\$101

Lodging by month (excluding taxes) | October 2020 - September 2021 (Sorted Data In Ascending Order): \$96, \$99, \$101, \$101, \$105, \$109, \$117, \$122, \$137, \$151

Table with Class Intervals (with Frequencies)

Primary Destination	County	March 2021 (x)	Frequency (f)	$f \cdot x$
Akron	Summit	\$99	1	\$99
Canton	Stark	\$96	1	\$96
Cincinnati	Hamilton/Clermont	\$151	1	\$151
Cleveland	Cuyahoga	\$137	1	\$137
Columbus	Franklin	\$122	1	\$122
Dayton/Fairborn	Greene/Montgomery	\$109	1	\$109
Hamilton	Butler/Warren	\$117	1	\$117
Mentor	Lake	\$105	1	\$105
Sandusky	Erie	\$101	1	\$101
Wooster	Wayne	\$101	1	\$101
			$\Sigma f = 10$	$\Sigma fx = 1138$

Mean

$$(ii) \bar{x} = \frac{\Sigma fx}{\Sigma f}$$

$$\bar{x} = \frac{\$1138}{10}$$

$$\bar{x} = \$113.80$$

The average price of the per diem for all 10 counties in March 2021 is approximately \$113.80.

Median

$$\Sigma f = 10$$

With this even-number sample size in ascending order, this means that the median will be between the 5th and 6th data values

List: ~~\$96, \$99, \$101, \$101, \$105, \$109, \$117, \$122, \$137, \$151~~

5th value (May 2021) = \$105

6th value (May 2021) = \$109

$$\$105 + \$109 = \$214$$

$$\frac{\$214}{2} = \$107$$

$$\text{Median} = \$107$$

In ascending order, the median price of the per diem for all 10 counties in ascending order (on March 2021) is \$107.

Mode

Mode is the data value that has the highest frequency/occurs most often.

The price that is charged the most often in the per diem as of March 2021 is \$101.

Midrange

Midrange: average of the minimum and maximum values

$$\text{Midrange} = \frac{\text{min} + \text{max}}{2}$$

$$\text{Min} = \$96$$

$$\text{Max} = \$151$$

$$\text{Midrange} = \frac{\$96 + \$151}{2}$$

$$\text{Midrange} = \$123.50$$

Canton (primary destination) had the lowest rate in March 2021 (*what are the possible reasons?*): \$96 – least expensive rate to Stark.

Cincinnati (primary destination) had the highest rate in March 2021 (*what are the possible reasons?*): \$151 – most expensive rate to Hamilton/Clermont.

The midrange, or average of the minimum and maximum rates, is \$123.50.

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$\text{Min} = \$96 \text{ (Canton)}$$

$$\text{Max} = \$151 \text{ (Cincinnati)}$$

$$\text{Range: } \$151 - \$96 = \$55$$

Canton had the least expensive per diem rate in March of 2021: \$96

Cincinnati had the most expensive per diem rate in March of 2021: \$151

The range of rates between Cincinnati and Canton (in ascending order) from March 2021 is \$55.

Variance (Sample)

Variance is a measurement that calculates the distance data values have from the mean, and it is equal to the square of standard deviation; sample variance affects a selection of data values).

$$\sum fx = 1138$$

$$\bar{x} = \$113.80$$

$$n = 12$$

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{\sum(x - 113.80)^2}{10 - 1}$$

$$s^2 = \frac{\sum(x - 113.80)^2}{9}$$

Rate (in Dollars)	Frequency	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
96	1	-17.8	316.84	316.84
99	1	-14.8	219.04	219.04
101	2	-12.8	163.84	327.68
105	1	-8.8	77.44	77.44
109	1	-4.8	23.04	23.04
117	1	3.2	10.24	10.24
122	1	8.2	67.24	67.24
137	1	23.2	538.24	538.24
151	1	37.2	1383.84	1383.84
$\Sigma f = 10$				$\Sigma f(x - \bar{x})^2 = 2963.60$

$$\sum(x - \bar{x})^2 = (316.84 + 219.04 + 327.68 + 77.44 + 23.04 + 10.24 + 67.24 + 538.24 + 1383.84) =$$

2963.60

$$s^2 = \frac{2963.60}{9} = 329.28888888888888888888888888889$$

The sample variance (or the variation of a sample/selection of data values) is approximately

329.29. This means that there is little to medium spread between the data values and the mean

(spread of the values from the mean)

Variance (Population)

Population variance is similar to the sample variance (calculates the distance of the data values from the mean), but it affects the whole dataset.

$$\mu^2 = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

$$\sum f(x - \bar{x})^2 = 2963.60$$

$$\mu^2 = \frac{2963.60}{10} = 296.36$$

The population variance (variance involving the entire population) is 296.36. This means that there is little to medium spread between the data values from the mean.

Standard Deviation (Sample)

The standard deviation is the measure of the distance the data values are from the mean. The bigger the number usually means more spread and variation between the values in the dataset.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{10-1}}$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{9}}$$

$$\sum (x - \bar{x})^2 = 2963.60$$

$$s = \sqrt{\frac{(2963.6)}{9}} = \sqrt{329.288888888888888888888888888889} = 18.146318879841413$$

The sample standard deviation (or a measure of variation of a selection of data values) is 18.146318879841413. This means that there is little spread between the values of the table/dataset from the mean.

Standard Deviation (Population)

Population standard deviation is similar to the sample standard deviation but it affects the entire dataset.

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}}$$

$$\sum f(x - \bar{x})^2 = 2963.60$$

$$s = \sqrt{\frac{2963.60}{10}} = \sqrt{296.36} = 17.2151096424042$$

The population standard deviation (affecting the entire dataset) is 17.2151096424042. This means there is little spread affecting the data values (entire population) of the dataset from the mean.

Measures of Location

The measures of location can be located after sorting the data into ascending order. Since there are 10 numbers, there are roughly 3 numbers in each quartile (quartiles are divided into 4 parts).

When the dataset is in ascending order and divided into four equal parts, the quartiles land on the 3rd data value (\$101), between the 5th and 6th data values (median, or \$107), and land on the 8th data value (\$122).

(Demonstration: (\$96, \$99, **\$101**, \$101, \$105), | (\$109, \$117, **\$122**, \$137, \$151)).

When split in half, it is easier to see where Q1 and Q3 would land (this is because \$101 and \$122 would be considered “medians”, or in the middle of their respective half).

First Quartile: \$96, \$99, (lands on) **\$101**

Second Quartile: (lands on) **\$101**, \$101, \$105

Third Quartile: \$109, \$117, (lands on) **\$122**

Fourth Quartile: (lands on) **\$122**, \$137, \$151

(The quartile land directly on \$101 and \$122, which are the “medians” of their respective dataset half (from the demonstration above))

Minimum value: \$96 (least expensive rate in the dataset/database, or minimum of the dataset)

Lower Quartile: \$101 (first quartile number in the dataset) $\frac{10}{4} = 2.5$ (25th Percentile) (to the 3rd

data value, which is \$101) When arranged in ascending order and divided into 4 equal parts,

\$101 is in the lower quartile, since it is in the middle (considered to be a “median”) of the first half between the minimum number and the actual middle (median).

Middle Quartile: \$107 (also known as the median of the dataset) $\frac{10}{2} = 5$ (50th Percentile =

50%); between 5th & 6th data value. When arranged in ascending order and divided into 4 equal parts, \$107 is in the middle quartile, since it is directly in the middle of the dataset (the actual median \$107).

Upper Quartile: \$122 (third quartile number in the dataset) $\frac{3}{4} * 10 = 7.5$ (75th Percentile) or to

the 8th data value, which is \$122; when arranged in ascending order and divided into 4 equal

parts, \$122 is in the upper quartile, since it is in the middle of the second half of the dataset (between the actual median and the maximum number).

Maximum value: \$151 (most expensive rate in the database/dataset, or maximum of the dataset)

Five-Number Summary

Min: \$96 (Canton to Stark) - (least expensive rate in the dataset/database, or minimum of the dataset)

Q1: \$101 (considered to be the “median” between the minimum number (\$96) and Q2 (between \$105 and \$109))

Q2 (Median): \$107

Q3: \$122 (considered to be the “median” (middle number) between Q2 and the maximum number (\$151))

Max: \$151 (Cincinnati to Hamilton/Clermont) - (most expensive rate in the database/dataset, or maximum of the dataset)

Conclusion

This project and database show the mean, median, mode range, midrange, variance, standard deviation, and quartiles of the data table. Seeing the different rates can help people compare the rates, and see which area works best for them, whether for money reasons, or for location convenience. This could help the per diem determine if they need to adjust their rates, or if they need to add more transfers/locations for the lodging based on the data. It also helps in which counties are best to transfer to, depending on money and location. This dataset is supposed to represent a month of lodging rates for ten counties of Ohio. With the transfers and counties, they are not really varied by much, but because these rates are smaller-sized, they are more manageable, especially with Canton to Stark being the least expensive and Cincinnati to Hamilton/Clermont being the most expensive.

Works Cited (Citations)

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